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THESE PLANS HAVE BEEN REVIEWED
FOR COMPLIANCE WITH
THE CURRENT EDITION
OF THE FLORIDA BUILDING CODE

Date: 11/15/19

Permit Number: BLD19-06035

BATSON-COOK CONSTRUCTION

Report of Supplemental Geotechnical Exploration and Engineering Services Proposed Publix Corporate Building Expansion Project Lakeland, Polk County, Florida

File No.: 19-51-9050

Ardaman & Associates, Inc.

CORPORATE HEADQUARTERS

8008 S. Orange Avenue, Orlando, FL 32809 - Phone: (407) 855-3860 Fax: (407) 859-8121

Branch Office Locations

Florida: Bartow, Cocoa, Fort Myers, Miami, Orlando, Port St. Lucie, Sarasota, Tallahassee, Tampa, West Palm Beach
Louisiana: Baton Rouge, Monroe, New Orleans, Shreveport

MEMBERS:

ASTM International
American Concrete Institute
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Ardaman & Associates, Inc.

Geotechnical, Environmental and
Materials Consultants

April 9, 2019
File Number: 19-51-9050

Mr. Donald W. Farris, Project Executive
Batson-Cook Construction
1101 Channelside Drive, Suite 401
Tampa, FL 33602

Subject: Report of Supplemental Geotechnical Exploration and Engineering Services
Publix Corporate Building Expansion Project
3300 Publix Corporate Parkway, Lakeland, Polk County, FL

Mr. Farris:

As per your authorization, we completed supplemental field exploration work to the north and south sides of the previously-conducted study area (refer Ardaman File No. 17-51-9021, dated March 27, 2017) for the proposed Publix Corporate Building Expansion Project at the subject site shown on Figure 1. Project information and scope of work were provided by you during a site visit by Binod Chalise, P.E. of Ardaman on March 26 and April 3, 2019. Based on the available information and our understanding of the project which comprises a three-story building addition on the east side of existing corporate building, the proposed expansion building's footprint could potentially be extended by approximately 10 feet on the north side and/or 20 feet or more on the south side in comparison to the width of the existing building at the ground level. While doing so, the building columns along the north line may move closer to the existing storm water pond bank.

The purpose of this exploration was to perform confirmatory borings on the north and south sides and determine if the areas were subjected to a construction process known as Deep Dynamic Compaction (DDC) which densified the upper fill and natural soil layers during site improvement work in 2000 and 2001.

FIELD EXPLORATION

Soil Borings

Our field operations consisted of conducting nine Standard Penetration Test (SPT) borings; B-6A, B-8A, and B-10A on the north side at the edge of the pond bank; B-1B, B-3B, and B-5B on the south side of the potential edge of the expansion area, at 20-foot offset from the projection of the south edge of the existing building at ground level; and B-1C, B-3C, and B-5C on the south side of the potential edge of the expansion area at 40-foot offset from south edge of the existing building at ground level, using procedures similar to those outlined in ASTM D 1586. A summary of the drilling and testing procedures utilized in the borings is included in the attached Appendix I. The borings were performed at the locations indicated on the attached Figure 1, to determine the stratification and estimate engineering properties of the subsurface soils beneath the proposed building site. The test boring locations were specified by you. Each borehole was terminated at a depth of 20 feet below the existing ground surface as specified by our firm. A continuous drilling and sampling procedure was performed within the depth of 10.5 feet, and then at 5-foot intervals thereafter to the termination depth. Upon completion, each borehole was plugged with a bentonite chip product.

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The borings were located in the field by our staff according to a boring plan, and by using hand-held GPS unit and/or tape measurements from known fixed locations. Land surface elevations at the soil boring locations were not determined. We understand that the third-party surveyor retained by you (AECOM) conducted a survey of some of the boring locations.

LABORATORY TESTING

Visual Review of Disturbed Soil Samples

The field soil boring logs and recovered soil samples were transported to our Bartow office from the project site. Each soil sample was examined by our staff to identify the engineering classification of the soil samples retrieved in the field exploration. The visual classification of the samples was performed in accordance with the current Unified Soil Classification System (ASTM D 2487) method for classification of soils for engineering purposes.

Moisture Content and Percent Fines Tests

Nine (9) disturbed soil samples were selected during the visual examination operations and tested to determine their moisture content, using test procedures specified in ASTM Standard D 2216. All of the moisture content test samples were retained following those tests, and then tested to determine the percentage of soil that passed through a U.S. No. 200 sieve (percent fines content), using the soil washing procedures described in ASTM Standard D 1140. The results of these tests are documented in Table 1 as well as in the attached soil boring profiles (see Figure 2), each plotted at the position where the sample was retrieved within the soil borings. The results of these tests are useful in confirming the Engineer's visual classification of the soil.

EVALUATION AND RECOMMENDATIONS

Site and Soil Evaluation

Foundation Conditions at North Zone

The three SPT borings (B-6A, B-8A, B-10A) conducted along the edge of the pond bank generally encountered loose sand to silty sand to a depth of 7 to 10 feet below land surface (bls), followed by loose to medium-dense, sand with silt to silty sand and clayey sand to a termination depth of 20 feet (bls). When compared to the soil data obtained from previously conducted borings (B-6, B-8, and B-10) along the north line, the upper foundation soils near the edge of the pond bank are relatively loose. The water table level was measured at depths ranging from 5 to 5.5 feet below land surface at the test borings (typically at or slightly higher than the pond level).

The north potential building expansion zone is shown on Figure 3. Based on the soil boring data, the upper foundation soils at north zone near the edge of the pond bank consists of approximately 6 to 8 feet of loosely-compacted fill. We believe that the north zone was not covered by Deep Dynamic Compaction (DDC) during the preparation phase of the future expansion pad during 2001.

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Foundation Conditions at South Zone

SPT borings (B-1B, B-3B, B-5B and B-1C, B-3C, and B-5C) were conducted on the south side at 20 and 40-foot offsets, respectively, from the projection of the south edge of the existing building at ground level. These borings generally encountered loose to medium-dense, sand to silty sand and clayey sand to a termination depth of 20 feet (bls). When compared to the soil data obtained from previously-conducted borings (B-1, B-3, and B-5) along the south line, the upper foundation soils at 20 and 40-foot offsets are comparable with the exception of Borings B-3B and B-3C which encountered relatively loose soil layers at a depth of 6 to 10 feet below land surface. The water table level was measured at depths ranging from 4.5 to 5.5 feet below land surface at the test borings.

The south potential building expansion zone is shown on Figure 3. Based on the soil boring data, the upper foundation soils at south zone up to 40-foot offset of south edge of the existing building consist of approximately 6 to 9 feet of well-compacted fill. We believe the south zone was prepared either by Deep Dynamic Compaction (DDC) or other method (i.e., placement of soil layers in lifts and compaction) during the preparation phase of the pad for future expansion pad during 2001.

Design Ground Water Table (North and South Zone)

The water table level was measured at depths ranging from 4.5 to 5.5 feet below land surface at the test borings. The storm water pond level on north side of the proposed site appeared to be approximately 6 feet below top of bank at the time of our exploration. Seasonal high ground water table information was not available. We recommend that the design high water table level for the proposed building site be considered at 3 feet below existing land surface for analysis and design purposes.

Construction Dewatering

Dewatering should be anticipated during construction for any excavation that is at or greater than 4 to 5 feet below land surface during the dry season and below 3 feet during the wet season. Control of ground water is required to achieve the necessary stripping, excavation, proofrolling, filling, compaction, and any other earthwork, site work, and/or foundation subgrade preparation operations required for the project, the actual method(s) of dewatering should be determined by the Contractor. Dewatering should be performed to lower the ground water level to depths that are adequately below excavations and compaction surfaces. Adequate ground water level depths below excavations and compaction surfaces vary depending on soil type and construction method, and are usually 2 feet or deeper. Dewatering solely with sump pumps may not achieve the desired results.

<u>Building Site Preparation Recommendations</u>

Refer to Pages 7 and 8 in our previous study report (Ardaman File No. 17-51-9021, dated March 27, 2017).

Building Foundation Design Recommendations

Following preparation of the subgrade soils as described in the previous study report, the shallow foundations for the building may be proportioned using the recommended allowable soil bearing pressure in the following table, to support the design dead load plus live load.

Building Walls	Allowable Soil Bearing Pressure (PSF)	Min. Footing Depth (Feet)	Individual Footing Least Width - Ft.	Continuous Footing Width - Ft.
North Perimeter Foundations * (adjacent to the storm water pond along the North line)	3,000	3	≥ 3 < 14	≥ 2 < 3.5
All other Foundations (Interior & South Perimeter along the South Line)	4,000	2	≥ 3 < 14	≥ 2 < 3.5
North Zone * (adjacent to edge of the pond bank) CONDITION: IF MOVED NORTH	2,500	3 (Option 1) 4.5 (Option 2)	≥ 3 < 14	≥ 2 < 3.5
South Zone (Up to 40-foot offset) CONDITION: IF MOVED SOUTH	3,000	2	≥ 3 < 14	≥ 2 < 3.5

* *Strapped to each other along the north line using reinforced concrete tie beams on-grade.*

A minimum soil cover of 24 inches, as measured from the bottom of the foundation system to lowest adjacent finished grade, should be provided for interior and south and east perimeter footings. The embedment of north perimeter footings should be 36 inches (increase to 54 inches if moved north to north zone using Option 2 as shown in Exhibit 1) below lowest finished adjacent grade. Isolated spread (column) footings should be proportioned to be at least three feet wide. A minimum lateral dimension of 24 inches should be provided, when proportioning the continuous foundation elements (wall footings). For ground floor concrete slab-on-grade, a slightly reduced modulus of subgrade reaction (K_{v1}) of 40 and 50 pci (pounds per cubic inch) may be used for North and South zones, respectively.

Predicted Performance of Shallow Spread Footings

Selection of the recommended soil bearing pressure was based primarily on considerations of limiting the expected settlement to tolerable values for the type of structure. Based on the expected magnitude of the foundation loads, we estimate a continuous wall footing total settlement of up to one inch for a footing width of up to 3.5 feet. We estimate the total settlement of individual pad foundations to be up to one inch for a pad least width dimension of up to 14 feet. It is our opinion that the settlement will occur incrementally as the dead weight loads are applied due to the predominantly granular nature of the foundation soils.

Batson-Cook Construction
File Number: 19-51-9050

CLOSURE

We appreciate the opportunity to be of service to you on this project. If there are any questions in regard to this report or when we may be of further assistance, please contact the undersigned at (863) 533-0858.

Sincerely,
ARDAMAN & ASSOCIATES, INC.
Florida Certificate of Authorization No. 5950

THOMAS J. LETO
No. 12458
STATE OF FLORIDA
PROFESSIONAL ENGINEER

4/3/19

Thomas J. Leto, P.E.
Senior Consultant
Florida License No. 12458

Binod R. Chalise, P.E.
Project Engineer

Brian D. Runkles, P.E.
Branch Manager

TJL/BRC/BDR:tc
Enclosures
Client Copies: (2)
File Copy: (1)

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TABLE 1

LABORATORY TEST RESULTS SUMMARY

SUPPLEMENTAL SOIL EXPLORATION FOR PUBLIX CORPORATE OFFICE BUILDING EXPANSION
LAKELAND, POLK COUNTY, FLORIDA

Prepared Date: 4/8/19

Disturbed Samples **Project No.: 19-51-9050**

INDEX PROPERTIES						
Test Hole No.	**Sample No.**	**Depth Range BLS (Ft)**	**Soil Description**	**USCS**	**Moisture Content (%)**	**Fines Content [Passing US No. 200 Sieve] (%)**
B-1B	2	2.5 - 3.0	Silty Sand	SM	11.9	12.7
B-3B	9	19.5 - 20.0	Sand w/Silt	SP-SM	19.9	9.6
B-5B	5	7.0 - 7.5	Sand w/Silt	SP-SM	18.2	9.1
B-1C	2	2.5 - 3.0	Silty Sand	SM	10.0	17.9
B-3C	3	4.0 - 4.5	Sand w/Silt	SP-SM	15.2	9.6
B-5C	3	4.0 - 4.5	Clayey Sand	SC	20.8	24.1
B-6A	3	4.0 - 4.5	Sand w/Silt	SP-SM	16.5	9.7
B-8A	5	7.0 - 7.5	Sand w/Silt	SP-SM	11.6	9.1
B-10A	8	14.5 - 15.0	Clayey Sand	SC	16.5	19.7

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APPROXIMATE TEST BORING LOCATION
(SOURCE: GOOGLE IMAGE DATED 1-17-16)

LEGEND

SPT Boring Locations - Conducted for this Study (Each 20' deep)
- Three Borings @ South 20' Offset: B-1B, B-3B, B-5B
- Three Borings @ South 40' Offset: B-1C, B-3C, B-5C
- Three Borings @ North near the Pond Bank: B-6A, B-8A, B-10A

Previously Conducted SPT Boring Locations (File No. 17-51-9021)
- Six Borings (B-6, 10, 11, 12, 13, 14) are 50' deep borings &
- Eight Borings (B-1, 2, 3, 4, 5, 7, 8, 9) are 30' deep borings

SUPPLEMENTAL TEST BORING PLAN

Ardaman & Associates, Inc.
Geotechnical, Environmental and Materials Consultants
1525 Centennial Drive Bartow, Florida 33830
Tele. 863/533-0858 Fax 863/533-7325 CAN. 5950

BATSON–COOK CONSTRUCTION
PUBLIX CORPORATE BUILDING EXPANSION PROJECT
LAKELAND, POLK COUNTY, FLORIDA

DRAWN BY: BRO	CHECKED BY: TJL	DATE: 4–8–19
FILE NO. 19–51–9050	APPROVED BY: Thomas J. Leto, P.E.	FIGURE NO. 1

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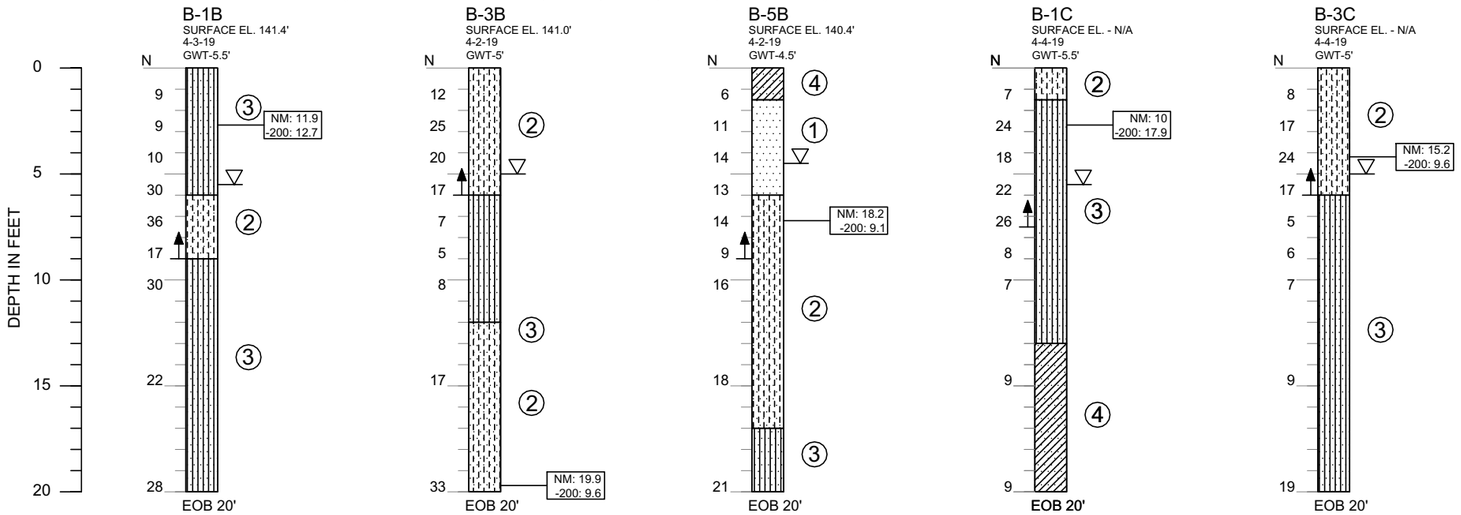
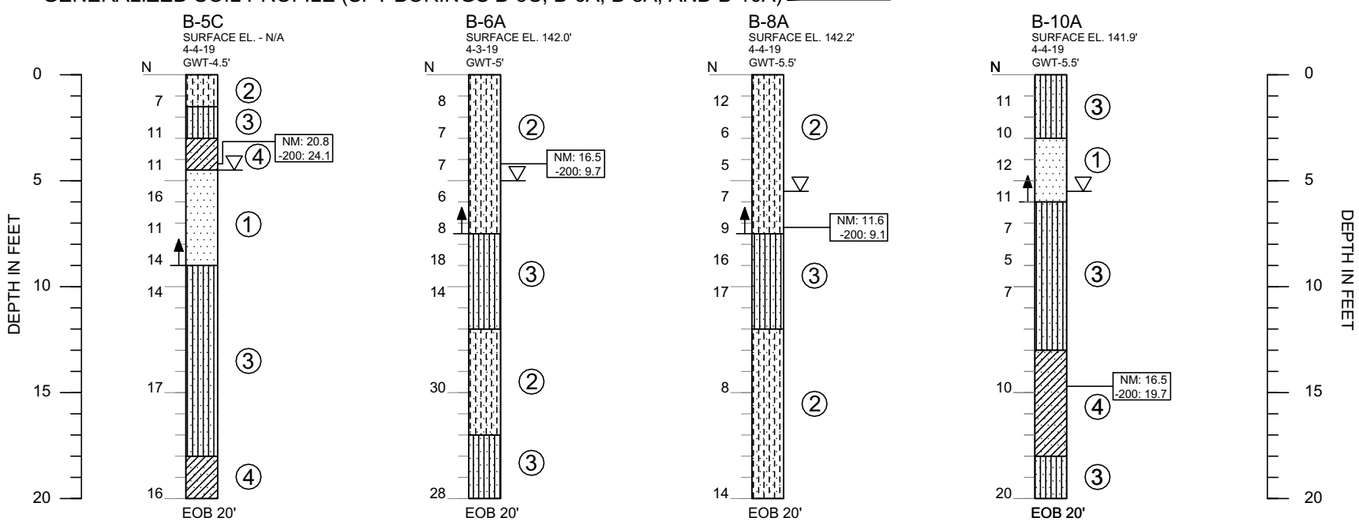
Permit Number: BLD19-06035

GENERALIZED SOIL PROFILE (SPT BORINGS B-1B, B-3B, B-5B, B-1C, AND B-3C)



GENERALIZED SOIL PROFILE (SPT BORINGS B-5C, B-6A, B-8A, AND B-10A)



Soils symbols	Description
1	Sand (SP); Light Brown or Light Gray
2	Sand with Silt (SP-SM); Brown to Dark Brown, or Light Gray to Gray
3	Silty sand (SM); Light Brown to Dark Brown, or Light Gray to Gray
4	Clayey sand (SC); Light Brown to Brown, or Light Gray to Gray, or Light Greenish Gray

Misc. symbols	Description
B-1B	Standard Penetration Test (SPT) Boring
N	Standard Penetration Resistance In Blows Per Foot
50/6"	Number Of Blows To Penetrate 6 Inches
HA	Hand Auger
▽	GWT: Ground Water Table Measured On The Date Indicated
SP,SM	Soil Classification Symbol ASTM 2487 Method For Classification Of Soils For Engineering Purposes
EOB	End of Boring
▲	Fill
NM	Natural Moisture Content (%)
-200	Percent Passing US No. 200 Sieve (%)

DRILLING NOTES:
DRILL RIG: CME-45
DRILLER: J. Mitchell
HOLE BACKFILL: Borehole Cuttings & Approved Materials
DRILL DATES: 4-2-19 through 4-4-19
DRILL METHOD: Rotary Wash with Drill Mud (SPT)
HAMMER: Auto Hammer
DRILL BIT: 2⅞ Tricone Roller or Drag Bit

ENGINEERING CLASSIFICATION

GRANULAR MATERIALS-RELATIVE DENSITY	SAFETY HAMMER SPT N-VALUE (BLOWS/FOOT)	AUTO HAMMER SPT N-VALUE (BLOWS/FOOT)
VERY LOOSE	LESS THAN 4	LESS THAN 3
LOOSE	4-10	3-8
MEDIUM OR COMPACT	10-30	8-24
DENSE	30-50	24-40
VERY DENSE	GREATER THAN 50	GREATER THAN 40
SILTS AND CLAYS CONSISTENCY	**SAFETY HAMMER SPT N-VALUE (BLOWS/FOOT)**	**AUTO HAMMER SPT N-VALUE (BLOWS/FOOT)**
VERY SOFT	LESS THAN 2	LESS THAN 1
SOFT	2-4	1-3
FIRM	4-8	3-6
STIFF	8-15	6-12
VERY STIFF	15-30	12-24
HARD	GREATER THAN 30	GREATER THAN 24

WHILE THE BORINGS ARE REPRESENTATIVE OF SUBSURFACE CONDITIONS AT THEIR RESPECTIVE LOCATIONS AND FOR THEIR RESPECTIVE VERTICAL REACHES, LOCAL VARIATIONS CHARACTERISTIC OF THE SUBSURFACE MATERIALS OF THE REGION ARE ANTICIPATED AND MAY BE ENCOUNTERED. THE BORING LOGS AND RELATED INFORMATION ARE BASED ON THE DRILLERS LOGS AND VISUAL EXAMINATION OF SELECTED SAMPLES IN THE LABORATORY. THE DELINEATION BETWEEN SOIL TYPES SHOWN ON THE LOGS IS APPROXIMATE AND THE DESCRIPTION REPRESENTS OUR INTERPRETATION OF SUBSURFACE CONDITIONS AT THE DESIGNATED BORING LOCATIONS ON THE PARTICULAR DATE DRILLED.

GROUNDWATER ELEVATIONS SHOWN ON THE BORING LOGS REPRESENT GROUNDWATER SURFACES ENCOUNTERED ON THE DATES SHOWN. FLUCTUATIONS IN WATER TABLE LEVELS SHOULD BE ANTICIPATED THROUGHOUT THE YEAR. ABSENCE OF WATER DATA ON CERTAIN BORINGS IMPLIES THAT NO GROUNDWATER DATA IS AVAILABLE, BUT DOES NOT NECESSARILY MEAN THAT GROUNDWATER WILL NOT BE ENCOUNTERED AT THOSE LOCATIONS OR WITHIN THE VERTICAL REACHES OF THESE BORINGS IN THE FUTURE.

NOTE

SURVEY DATA BY AECOM ON 4-2-19 WAS REFERENCED FOR SURFACE ELEVATION AT THE BORING LOCATIONS

SUPPLEMENTAL TEST BORING PROFILES

Ardaman & Associates, Inc.
Geotechnical, Environmental and Materials Consultants
1525 Centennial Drive Bartow, Florida 33830
Tele. 863/533-0858 Fax 863/533-7325 CAN. 5950

BATSON–COOK CONSTRUCTION
PUBLIX CORPORATE BUILDING EXPANSION PROJECT
LAKELAND, POLK COUNTY, FLORIDA

DRAWN BY: BRC	CHECKED BY: TJL	DATE: 4-8-19
FILE NO. 19-51-9050	APPROVED BY: Thomas J. Leto, P.E.	FIGURE NO. 2

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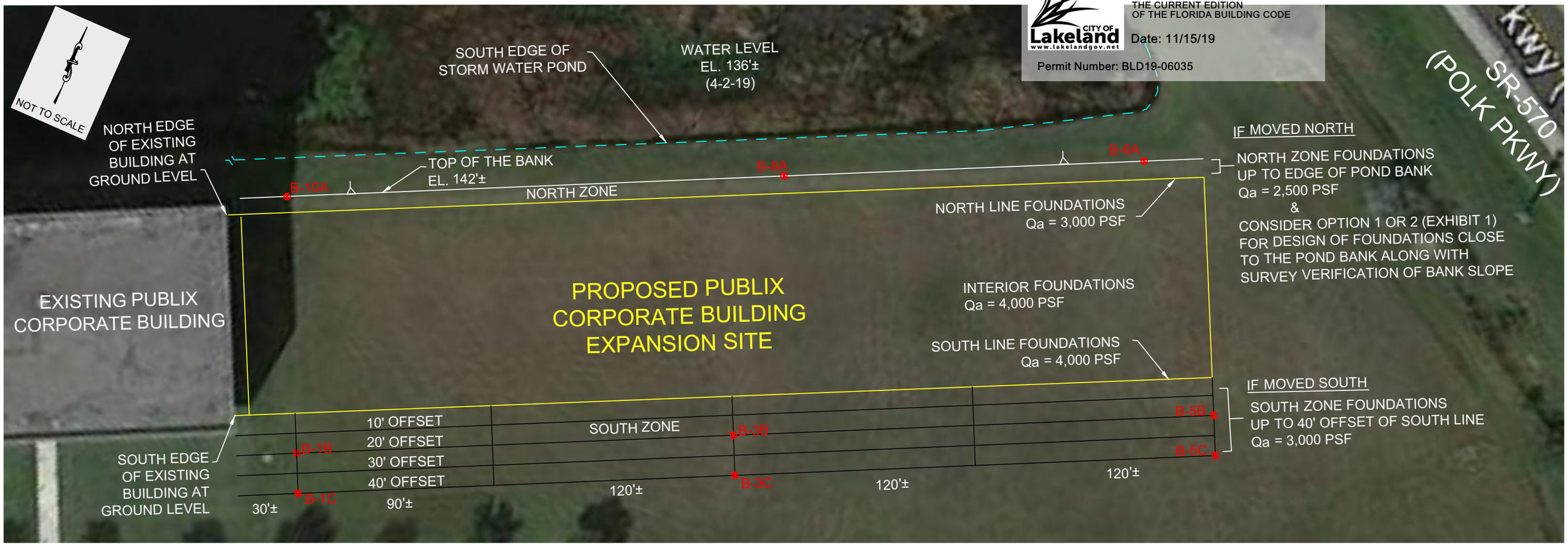

NOTES

Qa = ALLOWABLE SOIL BEARING PRESSURE (PSF)

MIN. DEPTH OF FOUNDATION IS 4.5 FEET FOR FOUNDATIONS ON NORTH ZONE (IF MOVED NORTH) USING OPTION 2 (SEE EXHIBIT 1) & IS 2 FEET FOR ALL OTHER FOUNDATIONS.

SURVEY DATA BY AECOM ON 4-2-19 WAS REFERENCED

ALLOWABLE SOIL BEARING PRESSURE FOR COLUMN FOUNDATION DESIGN

Ardaman & Associates, Inc.
Geotechnical, Environmental and Materials Consultants
1525 Centennial Drive Bartow, Florida 33830
Tele. 863/533-0858 Fax 863/533-7325 CAN. 5950

BATSON–COOK CONSTRUCTION
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EXHIBIT 1

FOUNDATION DESIGN OPTIONS AT
NORTH ZONE NEAR THE EDGE OF THE POND BANK

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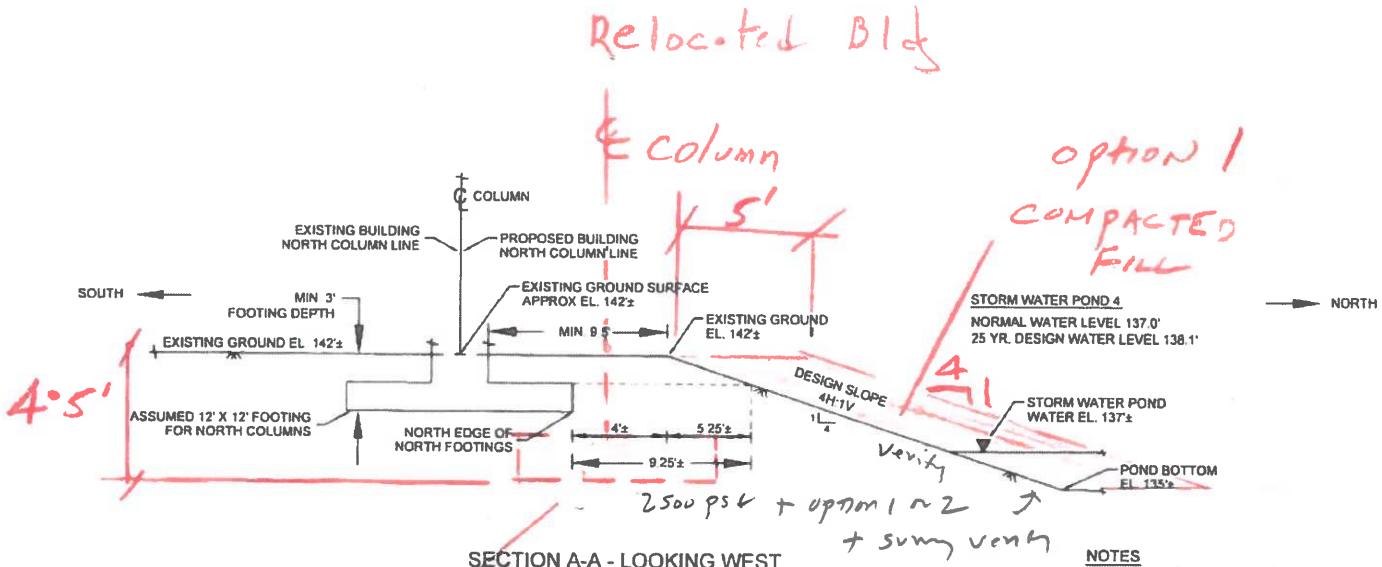

SECTION A-A - LOOKING WEST
NORTH COLUMN LINE OF PROPOSED BUILDING SAME AS EXISTING BUILDING
NOT TO SCALE

NOTES
(1) THE PLAN AND SECTION PROVIDED ARE FOR CONCEPTUAL DESIGN ONLY. ACTUAL SURVEY SHALL BE PERFORMED FOR CONSTRUCTION.
(2) PROTECT MARGIN & BANK WITH WELL-MAINTAINED SOD

option 2
Deepen Footing

SECTION DRAWING NORTH COLUMN LINE OF PROPOSED BUILDING SAME AS EXISTING BUILDING		
Ardaman & Associates, Inc. Geotechnical, Environmental and Materials Consultants 1923 Centennial Drive Bartow, Florida 33830 Tele 863/533-0858 Fax 863/533-7329 CAN 2650		
PUBLIX CORPORATE BUILDING EXPANSION PROJECT		
PUBLIX SUPERMARKETS, INC. LAKELAND POLK COUNTY, FLORIDA		
DRAWN BY:	CHECKED BY:	DATE: 3-27-17
FILE NO. 17-51-9021	APPROVED BY:	FIGURE NO. 8

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APPENDIX I

FIELD EXPLORATION PROCEDURES

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STANDARD PENETRATION TEST

The Standard Penetration Test is a widely accepted method of in-situ testing of foundation soils (ASTM D 1586). A 2-foot long, 2-inch outside diameter split-barrel ("spoon") sampler, attached to the end of drilling rods, is driven 18 inches into the ground by successive blows of a 140-pound hammer freely dropping 30 inches. The number of blows needed for each six inches of penetration is recorded. The sum of the blows required for penetration of the second and third, 6-inch increments of penetration constitutes the test result or N-value. After the test, the sampler is extracted from the ground and opened to allow visual examination and classification of the retained soil sample. The N-value has been empirically correlated with various soil properties allowing a conservative estimate of the behavior of soils under load.

The tests are usually conducted at 5-foot intervals. However, more frequent or continuous testing is done by our firm through depths where a more accurate definition of the soils is required. The test holes are advanced to the test elevations by rotary drilling with a cutting bit, using circulating fluid to remove the cuttings and hold the fine grains in suspension. Usually, the circulating fluid, which is a bentonite drilling mud, also serves to keep the hole open below the water table by maintaining an excess hydrostatic pressure inside the hole. In some soil deposits, particularly highly pervious ones, flush-coupled casing must be driven to just above the testing depth to keep the hole open and/or to prevent the loss of circulating fluid.

Representative split-spoon samples from soil at every five feet of drilled depth and from every different stratum are brought to our laboratory in airtight jars for further evaluation and testing, if necessary. After completion of a test boring, the hole is kept open if necessary, until a steady state ground water level is recorded. The hole is then sealed and backfilled.